

PDF Visual Accessibility Standard v2

Clear-print rules for study guides, workbook sheets, and practice resources.

Core rule

A learner should not have to zoom aggressively, distinguish faint colors, or fight dense layout just to read the material. Visual design must support the reasoning task, not compete with it.

1. Visual acuity and text size

- Body text: 14 pt by default. Avoid dropping below 12 pt in learner-facing PDFs.
- Questions and passage text: about 14-15 pt with comfortable line height.
- Headings: large enough to create an obvious hierarchy without using all caps as the main signal.
- For a separate modified large-print edition, use 18 pt or larger and simplify layout further.

2. Contrast sensitivity

- Use very dark text on a light background. Studo body text uses #172235 on white, far above the WCAG minimum contrast requirement.
- Use pale lavender as a background accent, not as low-contrast body text.

- Lavender text must use the darker accessible tone #51439C; do not use light lavender for small text.
- Avoid thin light-gray text, faint rules, watermark-like text, and low-opacity labels.

3. Color vision

- Never communicate correct/wrong, categories, importance, or instructions by color alone.
- Pair color with words, labels, bold weight, borders, symbols, or position.
- Keep the palette restrained so learners do not need to decode many colors.

4. Crowding, spacing, and visual field

- Use one clear reading column for passages and explanations.
- Keep generous spacing between questions and answer options.
- Use about 1.35 line spacing for the default clear-print edition.
- Avoid dense multi-column prose. Tables are reserved for true comparisons and should have roomy cells.
- Keep line length moderate; do not stretch paragraphs across the full page width.

5. Glare and backgrounds

- Default to white or near-white backgrounds with dark text.
- Use tinted callout boxes sparingly and only with strong text contrast.
- Avoid large saturated color fields behind long passages.

- Because glare preferences vary, the website should eventually offer alternate screen themes even though printable PDFs remain clear-print first.

6. Typography and recognition

- Use a plain sans-serif typeface with clear letterforms. Studo uses Inter for the PDF system.
- Prefer sentence case and title case. Long ALL-CAPS headings reduce word-shape cues.
- Use bold for short emphasis; avoid long italic or underlined passages.
- Keep answer labels A, B, C, D visually distinct and aligned.

7. Cognitive and scanning load

- One page should have an obvious starting point and predictable heading hierarchy.
- Use short paragraphs, lists, and worked examples rather than walls of text.
- Separate source passages from questions with spacing or a subtle neutral panel.
- Keep explanations direct: answer, evidence, reasoning, then a short mistake diagnosis when useful.

8. PDF accessibility beyond appearance

- Keep text as real selectable text, not screenshots of text.
- Use logical reading order and structured headings in the authoring document.

- Use real tables only when a table relationship exists.
- Check reflow/zoom behavior and run an accessibility audit when the publishing workflow supports it.

Research basis

- W3C WCAG 2.2: color must not be the only means of conveying information; normal text requires at least 4.5:1 contrast at Level AA, with 7:1 as enhanced contrast.
- RNIB Clear Print guidance: body text should ideally be 14 pt, use plain sans-serif fonts, and left-align text rather than fully justifying it.
- RNIB colour and contrast guidance: people with sight loss benefit from strong foreground/background contrast; very light text on pale tints can become effectively invisible.
- American Printing House for the Blind: large-print production requires deliberate choices about font face, size, color, spacing, and formatting; modified large-print materials commonly use 18 pt or larger.
- Adobe PDF accessibility guidance: accessible PDFs require logical structure/reading order and should support reflow and assistive reading.

Studo clear-print defaults

Element	Default
Body	14 pt Inter, dark ink
Passages / questions	14-15 pt, 1.35 line spacing
Primary headings	19-28 pt, bold
Body contrast	High contrast; dark ink on white

Accent	Dark lavender text + pale lavender background
Alignment	Left aligned; no justified prose
Tables	Only for genuine comparisons; roomy cells
Color meaning	Always duplicated by text/structure